

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B.Tech Examination (IT)

November 2013

IT302: Design and Analysis of Algorithm

Date: 22.11.2013, Friday

Time: 10:00 a.m to 1:00 p.m

Maximum Marks:70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

- (a) Explain at least two cases when divide and conquer technique is not preferred and why? 4
- (b) Why will we generally look at only WORST CASE running time of an algorithm? Explain with the help of a suitable example. 3

Q-2

- (a) How to analyze time and space complexity of algorithm? Explain with suitable example. 4
- (b) Define P, NP, NP complete and NP-Hard problems. 4
- (c) Generate at least 2-solution for 5-Queen's problem. Draw a pruned state space tree for the problem. 6

OR

Q-2

- (a) List and explain the significance of four factors that affects the efficiency of the backtracking algorithms. 4
- (b) What is Spanning tree? Give the Generic algorithm to find Spanning Tree for a given graph. List down similarities and differences between Kruskal's and Prim's algorithm. 5
- (c) Explain General Template for Divide and Conquer Technique with suitable example. 5

Q-3 Attempt Any Two.

- (a) Let 11 activities are given $S = \{p, q, r, s, t, u, v, w, x, y, z\}$ start and finished times for proposed activities are (0, 6), (6, 10), (3, 5), (1, 4), (3, 8), (5, 7), (2, 13), (5, 9), (8, 11), (8, 12), and (12, 14). Find the maximum size set of mutually compatible activities using Greedy Activity Selection algorithm 14
- (b) The array given as input to a quick sort algorithm is already sorted. What will be the running time of quick sort if the pivot is taken to be (i) the first element (ii) the last element and (iii) median of three(first, middle, last) elements? Give suitable example to support your answer.
- (c) List the factors that may influence the time and space complexity of a program. Write a recursive and non-recursive function to compute factorial of number. Compare the time requirements of non-recursive function with recursive version.

SECTION-II

Q-4

- (a) Define Binary Search. Explain How to solve binary search problem using divide and conquer technique. 4
- (b) What are the difference between greedy algorithm and dynamic programming? 3

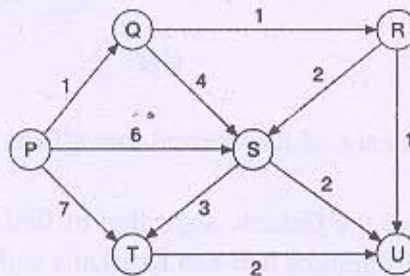
Q-5

- (a) Write an algorithm for naïve string matching. What is preprocessing and matching time for naïve string matching algorithm? 4
- (b) Solve following 0/1 Knapsack problem using Dynamic programming for five objects, whose weights are respectively 1,2,5,6 and 7 units and whose values are 1,6,18,22, and 28. The capacity of knapsack is 11 units. 6
- (c) Show that the following equalities are correct: 4
- (i) $5n^2 - 6n = \Theta(n^2)$
- (ii) $n! = O(n^n)$

OR

Q-5

- (a) Solve the following recurrence equation. 4
- $T_n = 2T_{n-1} + n,$ otherwise
 $\quad \quad \quad = 1$ if $n=0$
- (b) Dijkstra's single source shortest path algorithm on the following edge weighted directed graph with vertex P as the source is given below. 6



- (a) Output the sequence of vertices identified by the Dijkstra's algorithm for single source shortest path when the algorithm is started at node P.
- (b) Write down the sequence of vertices in the shortest path from P to U.
- (c) What is the cost of shortest path from P to U?
- (c) What is Articulation point in graph? Explain with the help of an example. 4

Q-6

Attempt Any Two.

- (a) Write the Insertion -SORT procedure to sort into non-increasing instead of non-decreasing order. Do the worst case and average case analysis. 14
- (b) Using Dynamic programming, find an optimum order of a matrix chain product whose sequence of dimensions is $\langle 5, 10, 3, 5, 6 \rangle$.
- (c) Consider $n=4, (w_1, w_2, w_3, w_4) = (2, 5, 10, 5)$ and $(p_1, p_2, p_3, p_4) = (40, 30, 50, 10)$ and capacity $W=16$. Find the optimal solution using **backtracking** for 0/1 knapsack problem using given data.